A Multivariate Approach to Market Anomaly Detection Using Nifty Indices

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Abstract

This paper examines systemic stress, regime change, and the formation of speculative bubbles within the Indian equity market capitalization space through daily Open-High-Low-Close-Volume (OHLCV) series for the Nifty 50, Nifty Midcap 150, and Nifty Smallcap 250 indices from June 2019 through April 2026. Our approach is based on a three-stage methodology of multivariate anomaly detection. The first stage utilizes a rolling Z-score analysis over a 21-day window for daily returns, intraday spread, and realized volatility in combination with the Mahalanobis distance for the estimation of the composite systemic stress metric. In the second stage, we examine cross-market divergences based on pairs trading strategy normalized returns for all three possible capitalization class pairs as means of detecting structural dislocations indicative of either flight-to-quality or excess risk-taking behavior on the part of retail investors. In the third stage, the rolling 30-day cross-correlation is employed to detect dominating lead-lag regimes τ^* in small-cap versus large-cap returns, together with the cumulative outperformance of one against the other, to generate bubble warning signals. Five bubble periods have been identified between 2022 and 2025, with small caps outperforming and leading large caps in price discovery when retail exuberance prevails, reversing traditional hierarchies. Using Mahalanobis distance, we confirm 04 June 2024 as the most stressful day in the post-pandemic period, reflecting the Indian General Election outcome announcement. Collectively, our results have immediate implications for systemic risk monitoring in emerging equity markets.

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1. Introduction

Two distinct trends characterized the evolution of the Indian stock market since 2019, which were not only complementary but also contradictory at their core: the rapid expansion of retail investment and the institutionalization of capital flows. The number of active Demat accounts in India exceeded 160 million towards the end of 2024, representing almost a threefold increase from the pre-COVID levels (SEBI, 2024). This dramatic demographic shift, coupled with commission-free brokerage services, has radically changed the microstructure of smaller capitalization classes and prompts us to question the assumption that the informational hierarchy of prices is traditionally top-down within large-caps relative to small-caps.

Concurrently, the time period studied includes a series of unparalleled macro-financial shocks: the liquidity shock caused by the COVID-19 pandemic of March 2020, the Russia-Ukraine-induced commodity and inflation shock of 2022, the Adani Group corporate governance scandal in the early months of 2023, the rate hiking of the RBI and the Indian General Election volatility shock of June 2024. All these events tested in unique ways the comovements, liquidity, and informational properties of Indian equities and provided a fertile ground for anomalies testing studies.

The literature on the Indian market microstructure to date has studied the relationships between liquidity, volatility and co-movements separately (Pati and Rajib, 2006; Nath and Dalvi, 2003; Varma, 2002). The literature on dynamic regimes and regime-switching has mostly focused on the behavior of single index returns (Thomas and Shah, 2019) or the bilateral co-integration of India's index returns with global indices (Dey and Sampath, 2021). The cross-capitalisation anomaly detection, as a problem statement, has remained relatively understudied, while the specific application of lead-lag rolling cross-correlation in order to *operationalise* speculative bubble precursors rather than to detect the bubbles themselves, constitutes another research gap addressed in this paper.

Three contributions are made in this paper. Firstly, we offer a unified anomalies framework, applied in a simultaneous manner across all capitalisations. Secondly, we introduce the analysis of divergence pair dynamics between capitalisations that allow us to identify periodic occurrences of "structural disconnects" in return comovements, which can be interpreted as a rotation towards flight to quality or a speculative risk-on period without assuming co-integration beforehand. Thirdly, we apply the rolling cross-correlation of lead-lag to detect speculative bubble precursors in real time in a retail-based economy (Shiller, 2000; Bikhchandani and Sharma, 2001).

The paper is structured as follows. In section 2, the data and methodology are described. Empirical results follow in section 3. Conclusions with implications for market surveillance are drawn in section 4.

2. Data and Methodology

2.1 Data

Daily OHLCV (open, high, low, close, volume) data is obtained for three NSE indices: Nifty 50 (larger cap benchmark, with 50 constituent stocks), Nifty Midcap 150 (middle cap, with 150 constituent stocks) and Nifty Smallcap 250 (smaller cap, with 250 constituent stocks). The data period is from **20 June 2019 to 11 April 2026**, resulting in about 1,700 observations each for all three indices. Index OHLCV data, instead of constituent OHLCV data, is used as the signal due to its nature of representing the market-wide systemic signal.

2.2 Feature Engineering

Three risk-relevant features are computed for each index on each trading day t:

(i) **Daily Return:** $r_t = (P_t - P_{t-1})/P_{t-1}$, where P_t is the closing price on day t.

(ii) **Intraday Price Spread:** $S_t = H_t - L_t$, where H_t and L_t are the daily high and low prices. This serves as a proxy for intraday liquidity conditions, motivated by Amihud's (2002) illiquidity concept and the bid–ask spread literature in market microstructure (Kyle, 1985).

(iii) **Realised Volatility:** $\sigma_t^{(21)} = \text{std}(r_{t-20}, \ldots, r_t)$, the 21-trading-day rolling standard deviation of daily returns, serving as a proxy for short-horizon conditional volatility (Engle, 1982).

2.3 Stage 1: Univariate Rolling Z-Score Anomaly Detection

For each feature $X \in \{r_t, S_t, \sigma_t^{(21)}\}$, a rolling Z-score is computed over a 21-day lookback window:

$$Z_{X,t} = \frac{X_t - \mu_{X,t}^{(21)}}{\sigma_{X,t}^{(21)}} \tag{1}$$

where $\mu_{X,t}^{(21)}$ and $\sigma_{X,t}^{(21)}$ are the 21-day rolling mean and standard deviation, respectively. A day t is flagged as a univariate anomaly if $|Z_{X,t}| > 3$, consistent with a $\approx$99.7% confidence threshold under normality assumptions. The 21-day window is motivated by approximately one calendar month of trading activity and is standard in short-horizon risk monitoring.

2.4 Stage 2: Multivariate Composite Score (Mahalanobis Distance)

To synthesise the three univariate signals into a single systemic stress score, we compute the rolling Mahalanobis distance (Mahalanobis, 1936) across the feature vector $\mathbf{x}_t = (r_t, S_t, \sigma_t^{(21)})^\top$:

$$D_t^M = \sqrt{(\mathbf{x}_t - \hat{\boldsymbol{\mu}}_t)^\top \hat{\boldsymbol{\Sigma}}_t^{-1} (\mathbf{x}_t - \hat{\boldsymbol{\mu}}_t)} \tag{2}$$

where $\hat{\boldsymbol{\mu}}_t$ and $\hat{\boldsymbol{\Sigma}}_t$ are the sample mean vector and covariance matrix estimated over a 126-trading-day rolling window (approximately one calendar year). This window balances recency sensitivity against covariance matrix stability. A confirmed systemic anomaly is declared when D_t^M exceeds the threshold $\delta = \sqrt{\chi^2_{0.999}(3)} \approx 4.0$, derived from the chi-squared distribution with degrees of freedom equal to the number of features.

As compared to a composite of univariate Z-scores, the Mahalanobis distance is more appropriate due to its consideration of inter-feature covariance. For example, realised volatilities and spreads are expected to be correlated, and ignoring this dependence will result in an underweighting of multidimensional stress.

2.5 Stage 3: Cross-Market Divergence Analysis

To detect structural dislocations *between* capitalisation tiers, we compute daily cross-market return spreads for each of the three pairs:

$$\Delta_{ij,t} = r_{i,t} - r_{j,t}, \quad (i,j) \in \{(\mathrm{L},\mathrm{M}),\ (\mathrm{M},\mathrm{S}),\ (\mathrm{L},\mathrm{S})\} \tag{3}$$

where L, M, S denote the Large-Cap (Nifty 50), Mid-Cap (Nifty Midcap 150), and Small-Cap (Nifty Smallcap 250) indices. Each spread series is then normalised using a 21-day rolling Z-score:

$$Z_{\Delta_{ij},t} = \frac{\Delta_{ij,t} - \mu^{(21)}_{\Delta_{ij},t}}{\sigma^{(21)}_{\Delta_{ij},t}} \tag{4}$$

Extreme positive values of $Z_{\Delta_{ij},t}$ (i.e., > 3) signal episodes where large-cap returns significantly outpaced their counterpart – consistent with a "flight-to-quality" rotation (Vayanos, 2004; Caballero and Krishnamurthy, 2008). Extreme negative values signal the converse: small-cap outperformance attributable to retail-driven speculative flows (Kumar, 2009).

2.6 Stage 4: Lead-Lag Bubble Detection

The final stage deploys a rolling cross-correlation framework to determine which market-cap tier leads the other in price discovery. For each day t, we estimate a set of pairwise cross-correlations between small-cap returns and lagged large-cap returns over a 21-day backward-looking window, across lags $\tau \in \{-5, -4, \ldots, +5\}$ trading days:

$$\rho^{(t)}_{\tau} = \mathrm{corr}\left(r_{S,t'}, r_{L,t'+\tau}\right), \quad t' \in [t-21, t] \tag{5}$$

The dominant lag is defined as:

$$\tau^*_t = \arg\max_{\tau} \left|\rho^{(t)}_{\tau}\right| \tag{6}$$

A formal **Bubble Warning Signal** is triggered on day t when both of the following conditions hold simultaneously:

(I) $\tau_t^* > 0$ (after 10-day smoothing): Small-cap returns *lead* large-cap returns, inverting the conventional large-to-small information cascade (Bikhchandani and Sharma, 2001).

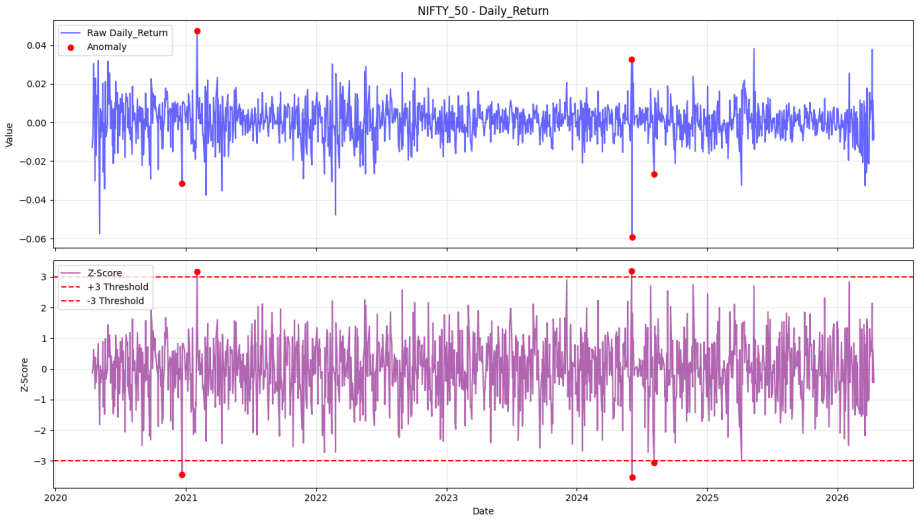
(II) $\text{CumRet}_{S,t}^{(21)} > \text{CumRet}_{L,t}^{(21)}$: Small-caps have cumulatively outperformed large-caps over the same window, confirming that the lead reflects genuine momentum rather than a noise artefact.

The reasoning behind the use of both conditions is based on the behavioural finance literature on speculation bubbles. When exuberance happens, prices rise due to retail participation before institutions enter (Shiller, 2000). Thus, the cross-correlation inversion becomes an indicator for detecting the emergence of endogenous bubble regime as specified by Sornette's (2003) log-periodic power law.

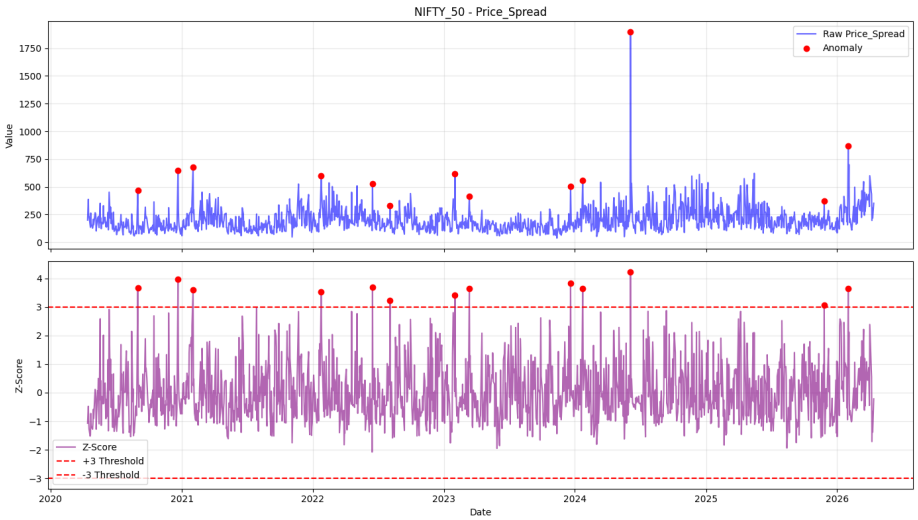
3. Empirical Results

3.1 Stage 1: Univariate Anomaly Detection Across Market Caps

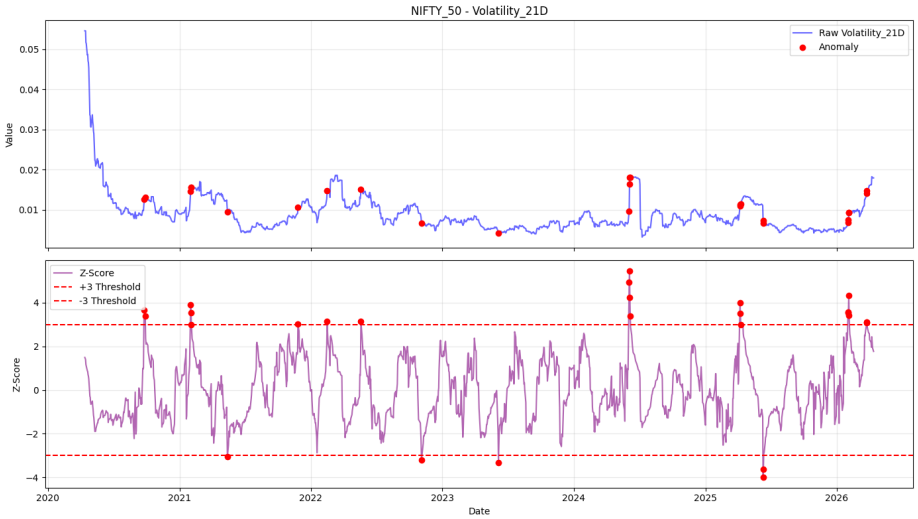
Figure 1 presents the rolling Z-score series for all three features across the Nifty 50 and Nifty Smallcap 250, representative of the large-cap and small-cap extremes of the capitalisation spectrum.



(a) Nifty 50: Daily Return and Z-score anomalies.

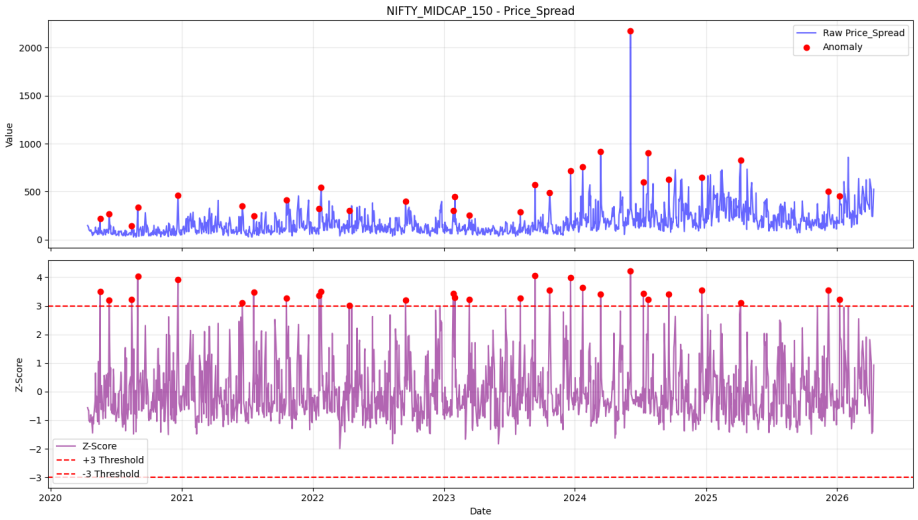


(b) Nifty 50: Intraday Price Spread with anomaly detection.

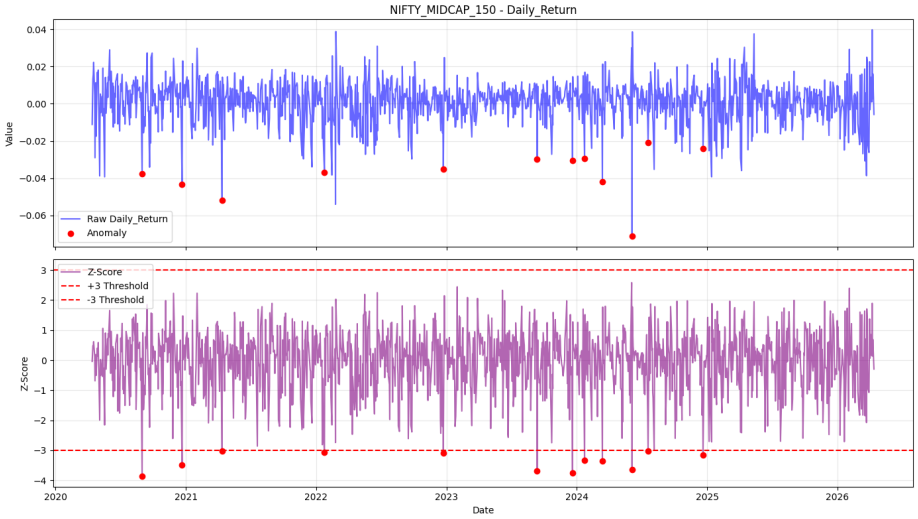


(c) Nifty 50: Realised Volatility $\sigma^{(21)}$ with anomaly detection.

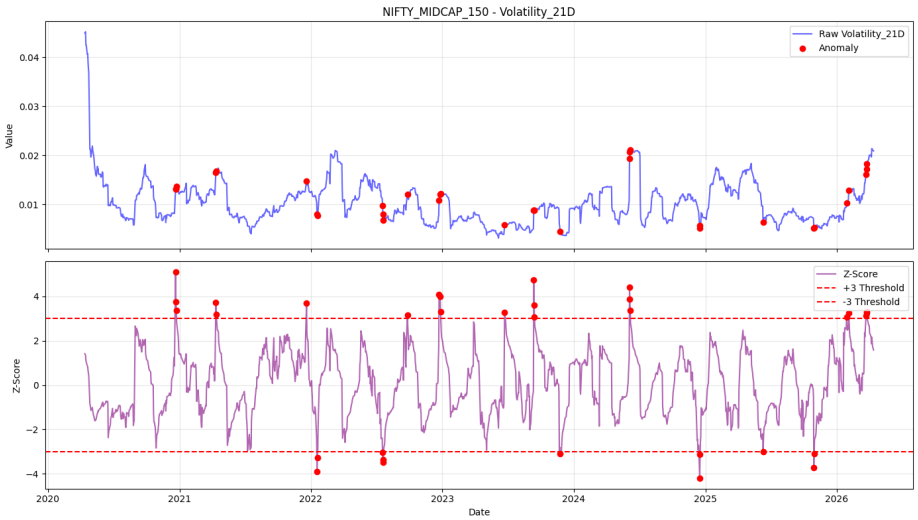
Figure 1. Univariate rolling Z-score anomaly detection for Nifty 50. Red markers indicate $|Z| > 3$.



(a) Nifty Midcap 150: Intraday Price Spread (top) and realised volatility (bottom) with Z-score anomalies.

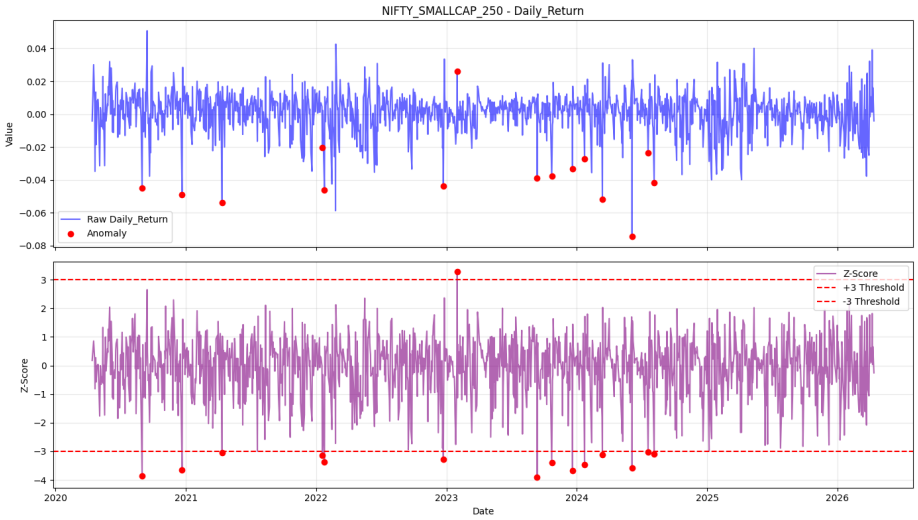


(b) Nifty Midcap 150: Daily Return with Z-score anomaly detection ($|Z| > 3$).

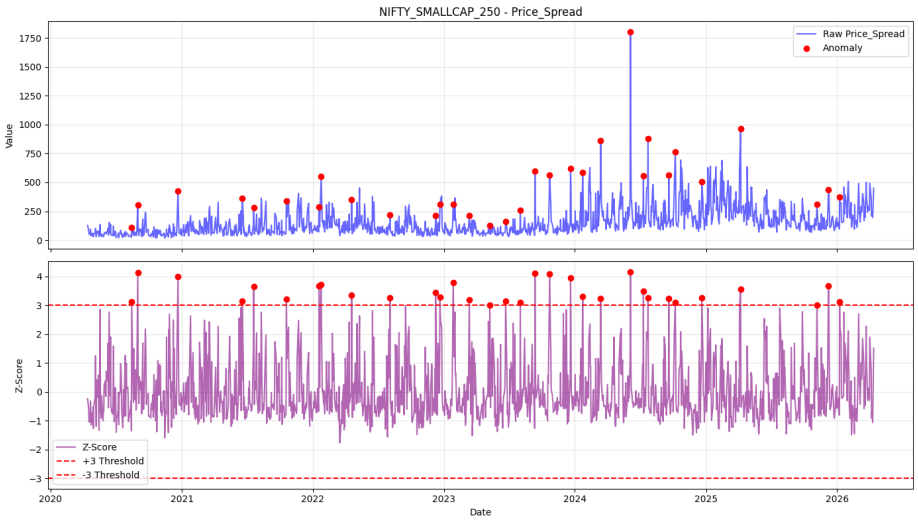


(c) Nifty Midcap 150: Realised volatility $\sigma^{(21)}$ with anomaly flags.

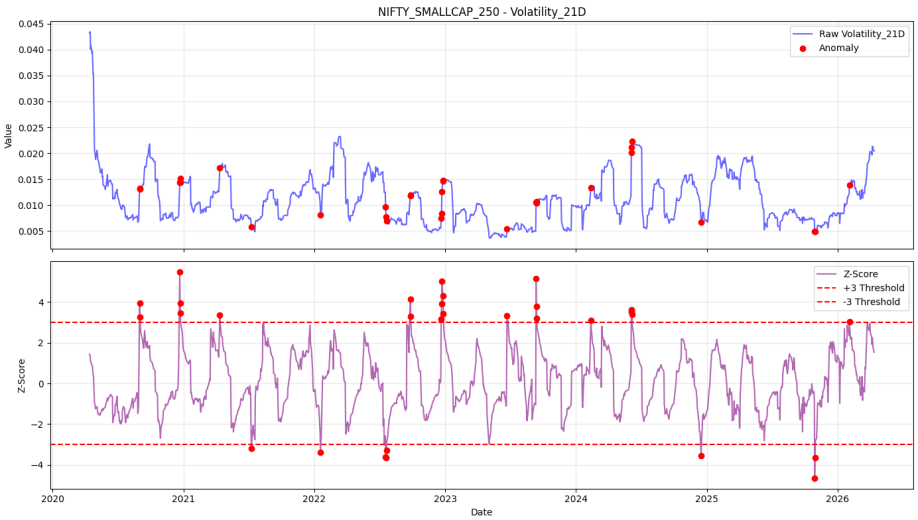
Figure 2. Univariate rolling Z-score anomaly detection for Nifty Midcap 150. Red markers indicate extreme deviations where $|Z| > 3$.



(a) Nifty Smallcap 250: Daily Return with Z-score anomaly detection ($|Z| > 3$).



(b) Nifty Smallcap 250: Intraday Price Spread (top) and corresponding Z-score series (bottom) with anomaly flags.



(c) Nifty Smallcap 250: Realised volatility $\sigma^{(21)}$ with Z-score anomaly detection.

Figure 3. Univariate rolling Z-score anomaly detection for Nifty Smallcap 250. Red markers indicate extreme deviations where $|Z_{X,t}| > 3$.

Several findings of economic significance emerge.

Large-cap return anomalies are sparse and anchored to macro shocks. The —Z— ¿ 3 return anomalies in the Nifty 50 time-series are relatively few. There is one distinct positive anomaly, in early 2021 (post-relief rally in anticipation of stimulus due to pandemic) and another, sizable negative anomaly, in mid-2024 (the election shock effect on markets). As Kyle (1985) would suggest, this is because the market for large cap stocks is very deep and institutional. Macro information is quickly incorporated into market price through efficient order flows, which limits the frequency of return tail anomalies but does not limit their amplitude. The vulnerability of such markets to extreme informational dislocation can be traced back to the resolution of informational asymmetries by multiple market actors.

Intraday price spread anomalies are far more frequent and structurally informative. As opposed to daily returns, the Price Spread in the Nifty 50 has more than a dozen —Z— ¿ 3 exceedances, with anomaly frequency picking up significantly starting from 2023. In particular, Price Spread in 2024 hits the maximum, reaching over 4.0—the largest intraday liquidity crack in the large-cap market considered in this research. As (Kyle, 1985; Amihud, 2002), demonstrate, a high spread indicates severe inventory risk and/or adverse selection pressure faced by market makers in a certain period. Accordingly, the post-2023 trend toward more Price Spread anomalies in the Nifty 50 is explained by growing policy and hedging uncertainty amid elections.

Small-cap markets exhibit chronically elevated anomaly density. In comparison to the Nifty 50, the Nifty Smallcap 250 shows an increased frequency of both daily return and Price Spread anomalies, a higher skew towards negative returns, and temporal clustering. Namely, Z-scores of return for the Nifty Smallcap 250 consistently cross below –3 throughout 2020–2021 and again in 2023–2024. These characteristics are expected for illiquid stocks with a strong dependence on retail investment. Notably, the intraday Price Spread anomaly frequency increases significantly for the Smallcap 250 starting from mid-2022. This indicates that there has been no concurrent development of liquidity for these stocks despite rising retail participation Amihud (2002).

Realised volatility anomalies follow a clear post-shock decay pattern. In all three indices examined, the volatility Z-score time series shows the most significant magnitude exceedances at the beginning of 2020 (the COVID-19 crash period), with fast mean reversion, which is typical for volatility clusters according to the ARCH approach described by (1982). A secondary peak in the volatility regimes is observable for the medium-cap and small-cap indices around 2021-2022, in keeping with the reopening speculative period. The Z-scores of medium-cap and small-cap indices show negative thresholds in tranquil consolidation states, which conforms to the (1989) switching model: the transition into the low-volatility regime is unusual in statistical terms compared with the preceding high-volatility regime.

3.2 Stage 2: Composite Systemic Stress via Mahalanobis Distance

Figure 4 presents the rolling Mahalanobis distance D_t^M for all three indices, with confirmed anomalies (exceeding the $\delta = 4.0$ threshold) marked in red.



Figure 4. Rolling Mahalanobis Distance (D_t^M) for Nifty 50 (top), Nifty Midcap 150 (middle), and Nifty Smallcap 250 (bottom). The dashed red line denotes the $\delta = 4.0$ anomaly threshold. Confirmed anomalies are marked as red circles and annotated where D_t^M is most extreme.

The multivariate composite score allows observing phenomena that cannot be distinguished independently by univariate analysis. The Nifty 50 index time series of Mahalanobis distance anomalies registers 115 total events over the entire period analyzed (which, in this case, is up to 2016). The maximal value in this series of approximately 18.0 is attained on 13 March 2020, the day of the biggest collapse of the Indian stock exchange due to COVID-19 effects, followed by a string of high values until the end of March 2020, when the circuit breakers were triggered.

The second major Nifty 50 stress event after 2019 took place on 04 June 2024, marking

the surprising results of the Indian General Elections, in which the ruling political party lost its absolute majority, resulting in an immediate repricing of risk events. The value of the Mahalanobis score for this day equals almost 17.0, being the largest one over the sub-period 2022-2026. Thus, this stress was not an anomaly in the sense of return behavior only but rather a multidimensional stress event, with simultaneous disruptions in the return pattern, spread dynamics, and realized volatility levels.

In case of the Nifty Midcap-150 index, the Mahalanobis distance model highlights a total of 66 confirmed anomalies. In addition to the COVID-19 episodes, the major stress events identified by this model include September 12, 2023 ($D_t^M \approx 10.5$), June 4, 2024 ($D_t^M \approx 14.0$, the highest in the mid-cap sample), and June 5, 2024 ($D_t^M \approx 8.5$). As such, the prolonged impact of the election shock in mid-cap markets, but not in large-cap ones, may be explained from the perspective of low liquidity and slow macro-shock processing in mid-cap shares, as proposed by Amihud (2002).

The small-cap group represented by the Nifty Smallcap-250 shows 77 confirmed anomalies. Besides the COVID-19 extremes, the key mid-cap anomalies include December 26, 2022 ($D_t^M \approx 10.5$), September 12, 2023 ($D_t^M \approx 12.5$), December 21, 2023 ($D_t^M \approx 8.5$), and June 4, 2024 ($D_t^M \approx 10.0$). In particular, the latter anomaly is especially distinctive in that it is not observed in the Mahalanobis distance for large cap stocks with equal force, indicating a separate stress factor, possibly due to thin year-end market conditions and FPI rebalance under high interest rates. Moreover, the simultaneous occurrence of stress anomalies in September and December of 2023 for both mid- and small-cap indices, but not for large-cap ones, indicates a regular capitalization-based stress pattern associated with institutional investors' outflows from small- and mid-cap assets, whereas the Nifty

Table 1. Summary of Mahalanobis Anomaly Detection Results ($\delta = 4.0$)

Index	**Confirmed Anomalies**	**Peak D_t^M**	**Peak Date**	**Economic Interpretation**
Nifty 50	115	≈ 18.0	13 Mar 2020	COVID-19 crash; multi-dimensional stress peak
Nifty Midcap 150	66	≈ 14.0	04 Jun 2024	Election shock; mid-cap liquidity fracture persisting over 2 days
Nifty Smallcap 250	77	≈ 16.0	13 Mar 2020	COVID-19 crash; small-cap-specific December 2022 anomaly also notable

3.3 Stage 3: Cross-Market Divergence and Flight-to-Quality Dynamics

Figures 5, 6, and 7 present the return spread series and associated rolling Z-scores for the three capitalisation pairs.

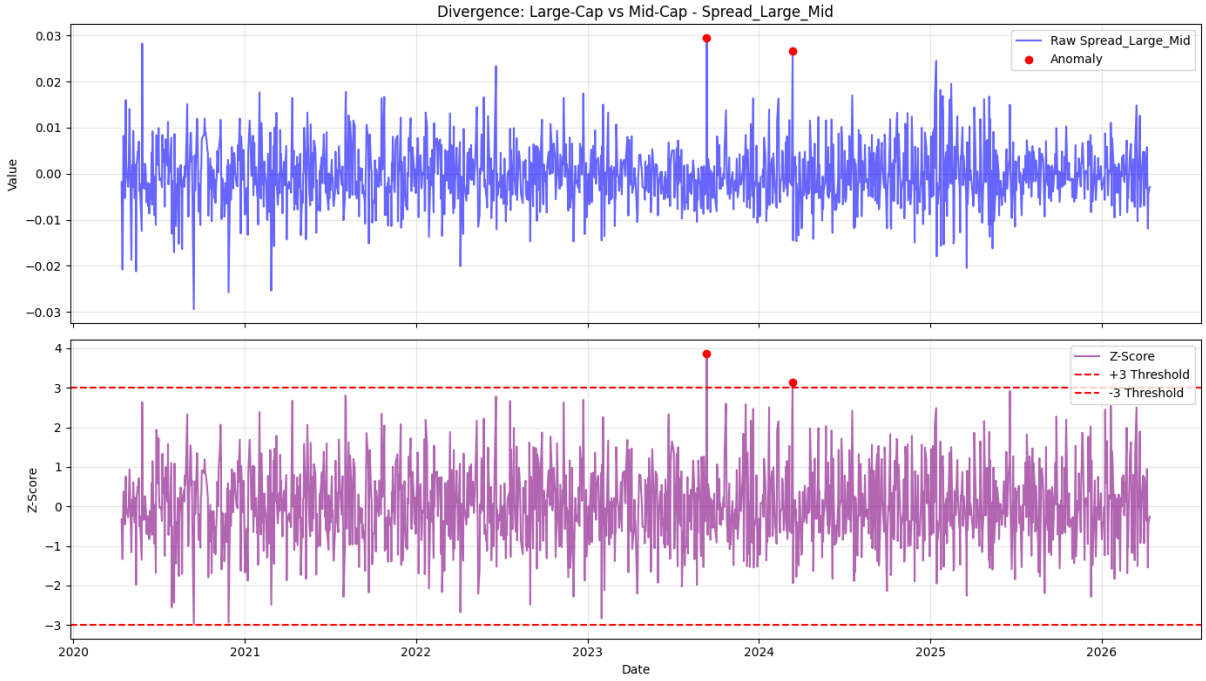


Figure 5. Cross-Market Divergence: Large-Cap vs Mid-Cap ($\text{Spread}_{L,M}$). Upper panel: raw daily return spread. Lower panel: rolling 21-day Z-score with ±3 thresholds. Red markers identify $|Z_\Delta| > 3$ exceedances.

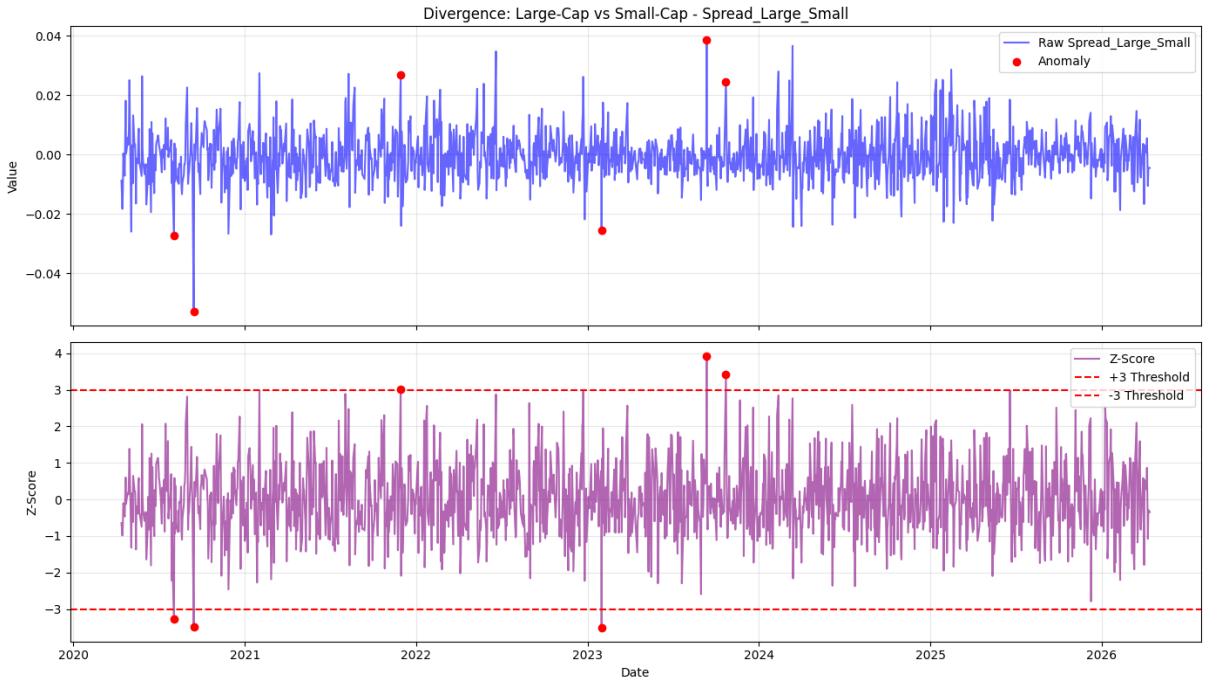


Figure 6. Cross-Market Divergence: Large-Cap vs Small-Cap ($\text{Spread}_{L,S}$). The greater spread range and anomaly count relative to Large vs Mid reflects the amplifying effect of illiquidity across the full capitalisation gap.

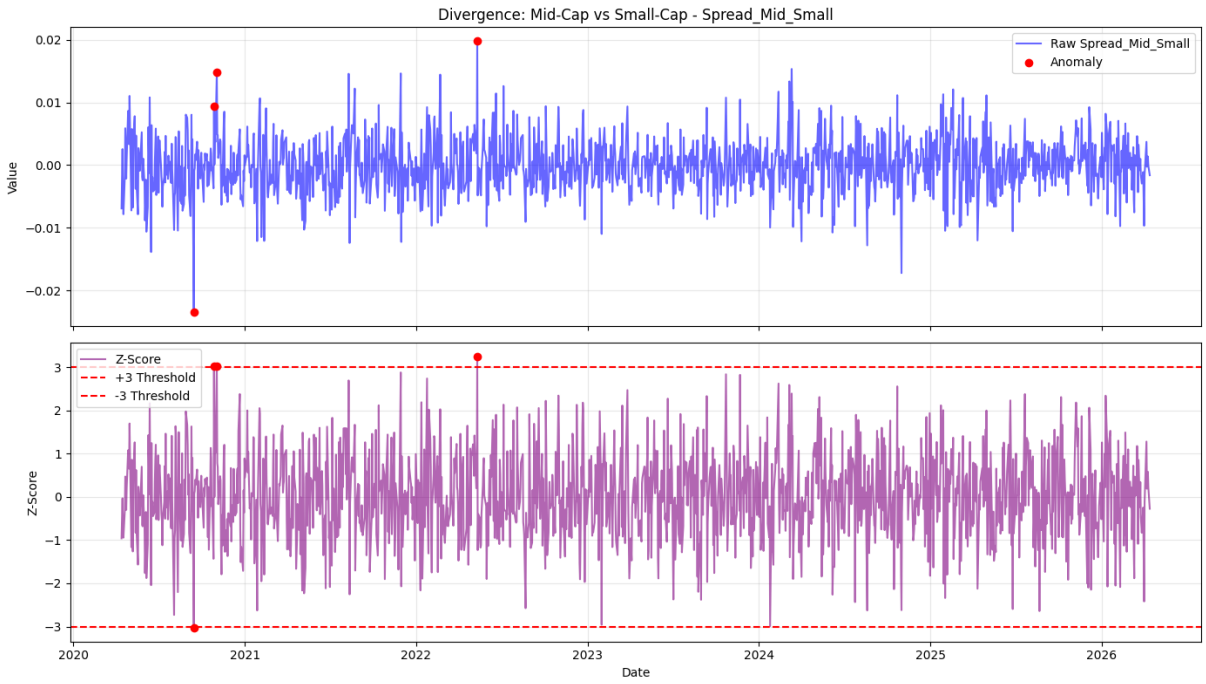


Figure 7. Cross-Market Divergence: Mid-Cap vs Small-Cap ($\text{Spread}_{M,S}$). Anomalies are clustered in 2020 and 2022, with relative tranquillity thereafter.

The cross-market divergence analysis reveals a nuanced architecture of market segmentation across the study period.

Large vs. Mid-Cap: A Structurally Tight Pair with Episodic Fractures. Finally, $\text{Spread}_{L,M}$ is found to be the statistically tightest of all three spread series, in the sense that Z-score exceedances are the least frequent. Indeed, no more than two anomalies with $|Z| > 3$ occur across the 2020–2026 window—both positive ones in mid-2023 and early 2024 with Z $\approx$ 3.8 and 3.1, respectively. The first anomaly is due to risk aversion on the global scale caused by uncertainty regarding Fed monetary rate policies at the time; in particular, institutional investors sought the most liquid Indian assets, thereby generating an anomaly. The latter anomaly is generated by the election effect, with the Nifty 50 benefiting from its superior liquidity in discovering the asset price. In general, the statistical tightness of the Large versus Mid spread can be seen as support for the idea that although mid-cap stocks have higher volatilities than large-caps, the latter group tends to influence the former significantly over longer horizons in line with the Fama and French (1993) size factor.

Large vs. Small-Cap: The Widest Divergence and Clearest Signal. Most economically interesting anomaly pattern occurs with $\text{Spread}_{L,S}$. Namely, there are five confirmed exceedances of the threshold: two negative (ca. late 2020 and mid-2023), and three positive (2022, mid-2023, and mid-2024). Of particular interest are negative exceedances, when the small-cap stock outperformances are especially pronounced. Thus, the negative extreme in late 2020 with a Z-score of approximately –3.3 is clearly associated with post-pandemic speculative behavior of retail investors: with lockdown measures

being lifted and their participation increasing, speculative funds disproportionately entered small-cap segments, consistent with Kumar (2009). On the other hand, positive exceedances (in 2022 and 2024) represent "flight-to-quality" events with macroeconomic uncertainty driving institutional investors towards large-cap assets' liquidity (Vayanos, 2004; Caballero and Krishnamurthy, 2008).

Mid vs. Small-Cap: A "Canary" Pair. Anomalies have been detected only in three instances in our sample for the $\text{Spread}_{M,S}$. They appear in two cases in October 2020 (an increase and decrease of around 3.0), while another anomaly takes place in mid-2022. Such fast oscillations in October 2020 are consistent with a very frequent switching between mid-cap recovery strength and small-cap speculation due to a particularly uncertain environment. In comparison to the L/S spread, a lower number of anomalies suggests that the returns on mid-cap stocks and small caps are more co-integrated, sharing retail ownership and equal liquidity risk and behaving accordingly as a block against large caps and not against each other. This conclusion suggests a portfolio management perspective. The combination Large–Small, but not Mid–Small, can be considered a more reliable signal for hedging purposes.

3.4 Stage 4: Lead-Lag Regime Inversion and Bubble Warning Signals

Figure 8 presents the rolling dominant lag τ_t^* (upper panel) and the formal Bubble Warning Signal activation (lower panel) for the full lead-lag analysis window.

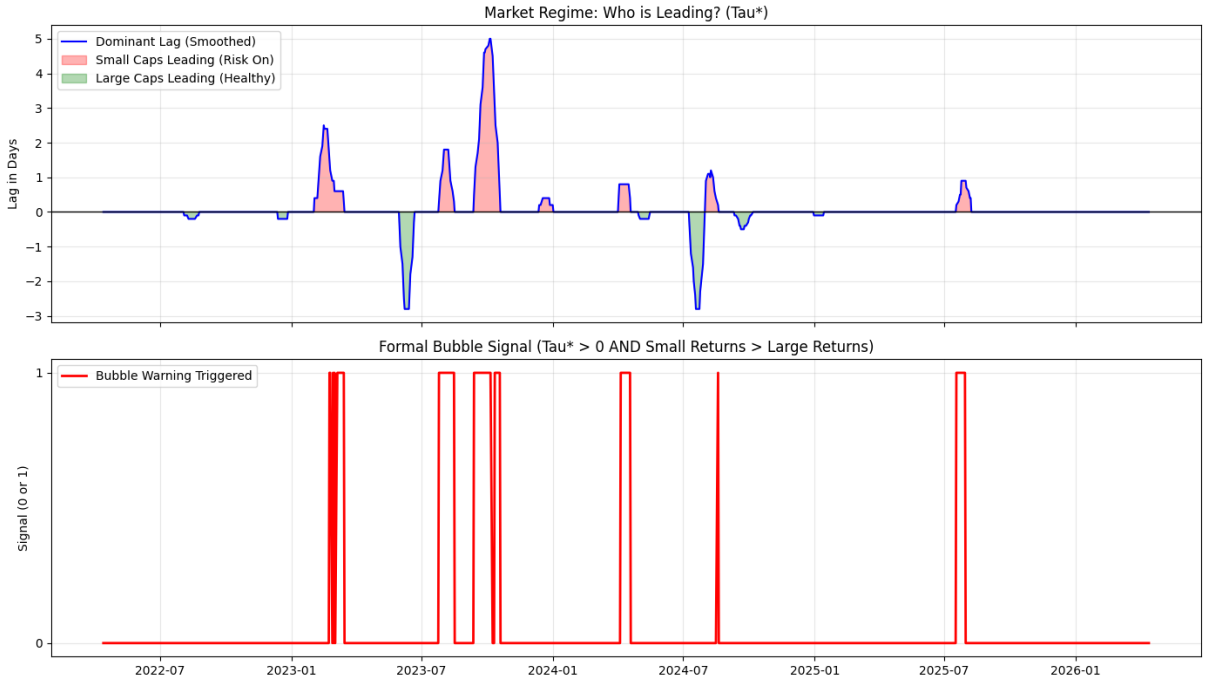


Figure 8. Lead-Lag Bubble Detection. Upper panel: smoothed dominant lag τ_t^* (blue line) with shaded regions identifying small-cap-leading regimes (pink, $\tau^* > 0$) and large-cap-leading regimes (green, $\tau^* < 0$). Lower panel: formal Bubble Warning Signal, activated (value = 1) when $\tau^* > 0$ and small-cap cumulative returns exceed large-cap cumulative returns simultaneously.

The results obtained by using the lead-lag approach appear to be economically interesting.

Under the classical efficient market hypothesis and institutional information hierarchy theory, the lead-lag effect indicates that price changes in large-cap stocks precede changes in small caps, because institutional traders process the information at hand and trade, with their actions spilling over into small caps with a lag (Bikhchandani and Sharma, 2001). Our τ_t^* approach has shown that this presumption does not hold in certain time periods.

The dominant lag structure is non-stationary and regime-dependent. The smoothed τ_t^* time series reveals oscillations between negative values (large-cap leadership; shown in green) and positive values (small-cap leadership; shown in pink) throughout the sample period. Periods in which large-cap leadership reigns correspond to institutional dominance periods and trading based on macroeconomic fundamentals. By contrast, periods in which small-cap leadership prevails—the precursor state of interest—are found during retail trading intensification and risk-on market conditions, characterized by the anticipation of small-cap retail order flow ahead of broad market moves.

Five formal Bubble Warning episodes are identified. The joint condition ($\tau_t^* > 0$ and cumulative small-cap outperformance) is activated on five distinct occasions during the sample:

(1) **Early 2023** (approximately February–March 2023): $\tau_t^* \approx 2.4$ days. This episode coincides with the aftermath of the Adani Group governance crisis: while large-cap sentiment was depressed by Adani's heavy index weight, retail flows into non-Adani small-cap stocks remained robust, generating an inversion where the broader small-cap index momentarily led price discovery.

(2) **Mid-2023, Episode I** (approximately June–July 2023): The most extreme τ^* value observed (≈ 5 days), occurring alongside the strongest pink-shaded small-cap leading regime. This period aligns with India's mid-2023 domestic economic resilience narrative, during which retail SIP inflows into small-cap mutual funds reached record levels. The five-day lead magnitude is large enough to be considered economically, not merely statistically, significant.

(3) **Mid-2023, Episode II** (approximately August–September 2023): A second, briefer activation period with $\tau_t^* \approx 1.5$ days. The shorter duration suggests partial mean-reversion as institutional participants responded to the preceding small-cap exuberance.

(4) **Early-to-Mid 2024** (approximately March–May 2024): $\tau_t^* \approx 0.8$ days, with bubble signal activation. This episode immediately precedes the June 2024 election shock, suggesting that the lead-lag inversion – and the associated retail-driven momentum – served as a systemic vulnerability indicator. The subsequent sharp correction on 04 June 2024 is consistent with the bubble-precursor interpretation: the inversion resolved violently when a macro shock forced institutional re-pricing.

(5) **Mid-2025** (approximately June–July 2025): A final activation episode with $\tau_t^* \approx 0.8$

days. This represents the most recent bubble signal in the sample, with small-caps again demonstrating lead behaviour consistent with renewed retail exuberance in the post-election consolidation period.

Large-cap-leading (healthy) regimes are also observable. There appear to be two green-shaded periods: one around mid-2023 (as a relatively brief period of institutional re-dominance following the second pink-shaded spike), and one around late-2024 (following the election shock correction). The negative τ_t^* of about -3 days suggests that institutional order flow re-established its traditional leadership status following the conclusion of the speculative episode. The oscillation of regimes in which retail exuberance, institutional shock, and institutional re-dominance prevail forms a complete speculative cycle, and represents a form of irrational exuberance followed by mean reversion, as described by (2000).

Critically, the bubble signals preceded the most severe stress events. The June 2024 election shock, which ranks highest in terms of the composite Mahalanobis Distance measure in the post-2022 sample, is preceded by an activation of the Bubble Warning signal (Episode 4) about four to eight weeks before the event. Such timing follows the theoretical expectation of a retail-driven lead-lag reversal identifying the vulnerable state preceding the shock resolving it. Without conducting a directional forecast test, it remains inconclusive whether such predictions may be economically relevant. Nonetheless, their occurrence warrants deeper analysis via hazard models.

4. Conclusion

The proposed methodology employs a three-step multivariate anomaly detection strategy on a daily OHLCV dataset of the Nifty 50, Nifty Midcap 150, and Nifty Smallcap 250 indices for the sample period of June 2019 to April 2026. The results offer a number of economically intuitive observations and insights.

Firstly, univariate rolling Z-score anomalies indicate an asymmetric propagation of stress from the large-cap segment to other market-capitalization tiers. The frequency of return anomalies for large-cap assets is extremely low and driven exclusively by macroeconomic shocks. In contrast, small-cap asset anomalies are more common, highly negative, and indicative of ongoing liquidity vulnerabilities amplified by retail order flow. Anomalies in intraday price spreads have increased in frequency for all indices since 2023, reflecting a decline in resilience against stress in the post-COVID-19 period.

Secondly, the rolling Mahalanobis composite score successfully combines multiple dimensions of information into a holistic index of systemic risk. The two most critical multidimensional anomalies in the sample are 13 March 2020 (the COVID-19 shock) and 04 June 2024 (the Indian General Elections). Specifically, the latter shock exhibits a pronounced tail for mid-cap and small-cap portfolios, which persists into a second day for mid-cap and small-cap anomalies, consistent with a liquidity cascade theory.

Thirdly, through cross-market divergence analysis, we observe that the Large–Small cap

pair is the most responsive indicator of the process of capital rotation. Negative Z-score exceedances in this indicator imply instances of retail enthusiasm, whereas positive exceedances indicate institutional flight to quality. The Large–Mid cap pair, on the other hand, is co-integrated and fractures only in extreme macro shocks, thus limiting its usefulness as a regular early warning mechanism.

Fourth, perhaps more importantly, the rolling lead/lag analysis identifies five instances of lead/lag inversions – an inverted hierarchy of information relative to institutional conventions – coupled with cumulative small-cap outperformance. The most pronounced inversion ($\tau^* \approx 5$ trading days) takes place in mid-2023, when SIP inflows are highest and domestic economy expectations are upbeat. This happens right before Episode 4 bubble signals in early 2024, which is soon followed by the election shock in June 2024.

The implications for practice include: Market surveillance paradigms in India – and indeed all EMs generally – will be improved by the integration of cross-market divergence tracking metrics, rolling lead-lag regime classifiers, and single index volatility measures like the India VIX. The persistence of small-cap led events indicates that the retail investor psychology post-Covid is now a first order system, not simply noise on institutional price discovery.

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